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Agentic Admission

A multi-agent LLM panel for MTE dossier evaluation at EPFL

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Abstract

The MTE admission committee at EPFL processes around 130 dossiers per semester and another wave of 100+ in the autumn round. Each dossier carries a transcript, two reference letters, a motivation letter, a CV, and a heterogeneous bundle of certificates. The committee meets for a few hours and decides. This project builds a decision-support system that emulates the committee's deliberation with four specialist LLM evaluators and a Chair agent, calibrated against the committee's own historical verdicts on 1,342 candidates from 2020 to 2026. The system normalises Excel inputs that change schema every year, anonymises 12,529 dossier documents with a Presidio + GLiNER pipeline that reaches 100% coverage on emails, phones, addresses, and Swiss identifiers (above 99% on names after two targeted fixes), runs a structured-output multi-agent deliberation against vLLM-served models, and applies a three-band GPA gate that mirrors the rule a former chair documented in 2026. On the Spring 2025–26 cohort the system reaches 82% agreement with the committee on the 28 EPFL-internal candidates (up from 68% before the band rule), with a 5-run majority-vote stability of 76.6% on the full cohort. The report tells that story end to end: the engineering decisions that made it reproducible, the academic decisions that made it falsifiable, what we would have done with more time, and what would have to change to port the system to other EPFL admission filières.

Keywords: LLM-as-judge, multi-agent deliberation, admission decision support, calibration, anonymisation, EPFL MTE

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1 Introduction

1.1 Problem

Every semester the MTE admission committee at EPFL receives around 130 applications in the spring round and a second wave of 100+ in the autumn round. A dossier is a heterogeneous package: an academic transcript on whatever scale the home institution uses (US 4.0, French 20, Swiss 6, German 1.0–6.0), two recommendation letters, a motivation letter, a CV, English-proficiency scores, and a stack of certificates. Committee members read all of this and meet for two afternoons to deliver an accept / reject / waitlist verdict for each candidate.

The bottleneck is not the decision itself, it is the reading. A reasonable reader needs 20–30 minutes to form an opinion on one dossier. At 130 dossiers per round, two committee members reading every file means 80–130 hours of reading split across a calendar that collapses near the deadline. Cases at the boundary get less time than they need; clear-accept and clear-reject cases get more time than they need. The committee has asked, for several years, whether a triage layer could pre-classify dossiers and surface only the borderline ones for full discussion.

This project is the technical answer to that question. The system does not decide; the committee still decides. The system reads every dossier, scores it on eight dimensions, recommends a verdict, and records its reasoning in a form the committee can audit.

1.2 Research question

> **Research question**
> Can a multi-agent panel of large language models, calibrated against the historical decisions of a real admission committee, reach an agreement level on a held-out cohort that makes the system useful as a triage tool without introducing systematic bias?

Three sub-questions sit underneath:

1. **Calibration.** How do we measure agreement when the ground truth is itself a noisy panel of humans, and what is a meaningful target?
2. **Stability.** LLMs are stochastic. How much does the same dossier move from run to run, and which scoring dimensions are stable enough to trust?
3. **Auditability.** The committee will not adopt a system whose decisions cannot be challenged in a meeting. What output format makes the system's reasoning easy to interrogate?

1.3 Contributions

This report describes four contributions:

1. An **extraction and anonymisation pipeline** that takes six years of inconsistent committee Excel exports and 12,529 dossier documents, and produces a single canonical Parquet of 1,342 candidates with 100% coverage on emails, addresses, IBANs, AVS, and 96.1% on names (see Section 3).
2. A **multi-agent deliberation architecture** with four specialist evaluator personas and one Chair, run under structured outputs, costed under $0.10 per dossier on Gemini Flash and $0.70 on Sonnet 4.6 (Section 4).
3. A **calibration loop** that compares the system against the committee's own decisions on Spring 2025–26 across five runs, with iterative prompt fixes that moved EPFL-internal agreement from 50% to 82% in three weeks (Sections 6 and 7).
4. A **spec-first engineering posture** with type checking, ruff `ALL`, atomic PRs, and a public changelog of 14 weekly reports. Every architectural change is traceable to a design document and a PR (Section 5).

1.4 Reading order

Sections 1–4 describe the system. Sections 5 and 6 split the work into its engineering and academic halves. Section 7 presents results. Sections 8 and 9 look forward: what we would do with more time, and what would change to port the system to a different EPFL programme. Appendices A–C carry the long-form artefacts (full evaluator prompt, per-cohort agreement table, file inventory).

2 Background

2.1 Why LLMs

Three properties of recent instruction-tuned LLMs make them a plausible fit for dossier reading. First, they handle long, mixed-form text without bespoke parsing. Second, they emit structured outputs that downstream code can consume. Third, they can be steered with a prompt that encodes a rubric the committee actually uses, including edge cases like the EPFL-internal three-band rule (Section 6).

None of this means the LLM is calibrated. The literature on LLM-as-judge is consistent on one point: zero-shot judgements drift from human judgements in ways that depend on the model, the prompt, the domain, and the ground-truth source [1], [2]. The remedy is to measure that drift on the target task and then design either the prompt, the rubric, or the aggregation to close the gap. That measure-and-close loop is the spine of this project.

2.2 Multi-agent deliberation

Single-agent scoring conflates evaluation criteria that have different evidential bases. A motivation letter and a transcript do not weigh the same kind of signal. We split the work into four specialist personas and a Chair, each with its own prompt and its own narrow scoring task. The Chair does two things: surface meaningful disagreement between specialists, and synthesise a final recommendation with explicit weights.

This is a classical separation-of-concerns argument applied to a prompt design. The benefit shows up in audit: when the system disagrees with the committee, we can read the four specialist scorecards side by side and ask which one is wrong. A single-agent score gives no such handle. Verga et al. report that a jury drawn from several model families beats a single larger judge [3]; our four personas share one model, a limitation the threats list in Section 6 carries as an open item.

2.3 Calibration against a noisy panel

The committee is the ground truth and the committee is noisy: the same dossier read by a different pair of committee members on a different day can produce a different verdict. Section 6 treats each recorded decision as a point estimate and derives the ceiling that noise puts on any agreement number.

2.4 Related work

The closest prior systems fall into three groups. First, the LLM-as-judge frameworks: MT-Bench [1], AlpacaEval [4], and JudgeBench [5] benchmark model responses against reference answers. The biases this literature documents in LLM judges (verbosity [6], position [7], and the wider taxonomy in [8]) apply to any system that scores free-form text; Section 6 carries the ones we have not yet measured as threats.

Second, admission systems trained on committee history. GRADE [9] predicted graduate-admission verdicts at UT Austin from past committee decisions and was retired in 2020 after public criticism that a model calibrated on historical decisions reproduces historical bias. The criticism transfers to this project unchanged, and it is the reason the bias audit ranks first among the open items in Section 8.

Third, multi-agent deliberation. Debate between LLM agents improves factuality on reasoning benchmarks [10], and it also degenerates: Liang et al. observe agents converging on their first answers instead of producing new arguments [11]. Our debate phase is conditional (it runs only on detected conflicts) and bounded at two rounds. The 5-run majority vote of Section 6 is self-consistency [12] applied to a three-way classification.

The regulatory frame is the EU AI Act, which classifies systems that determine access to education as high-risk [13]. The fairness literature and the Act point the same way: we treat the system as decision-support and not decision-making. Section 6 returns to the design choices that follow from that posture.

3 Data and anonymisation

3.1 Source corpus

The committee shares two artefacts per round. The **decision Excel** records one row per evaluator per candidate (so three rows per candidate for a triple-evaluator design), with the committee's final column appended at the right. The **dossier zip** contains one sub-directory per candidate with the transcript PDF, the two recommendation letters, the motivation letter, the CV, the English-test certificate, and an IS-Academia synthesis form. The schema of the Excel changes every year. The internal structure of the dossier zip changes per dossier.

Year	Excel sheets	Candidates	Notable schema change
2020–21	2	228	Triplet evaluators, no GPA column
2021–22	2	241	GPA column added, hidden gender col
2022–23	2	233	Decision column moved
2023–24	2	218	Reference grade scale changed
2024–25	2	194	Two new evaluators rotated in
2025–26	2	228	Section column renamed

Table 1: Six years of decision Excels, 1,342 candidates total. Each row encodes a vote; the parser groups rows back into candidates and detects evaluator triplets.

3.2 Extraction pipeline

The extraction module (`src/agentic_admission/extraction/pipeline.py`, 523 lines) handles the year-to-year drift with three layers:

1. **Fuzzy header matching.** A header rule maps a regular-expression pattern to a canonical column name. The same logical column appears as `GPA / Average`, `Note moyenne`, `Average grade`, or `Moy. / Note` across years.
2. **Multi-row grouping.** The committee writes one row per evaluator. The grouper detects triplets by family-name + date proximity and collapses them into a single `CandidateRecord` with an evaluator list.
3. **Hidden-column detection.** Some years store the candidate's gender in a hidden Excel column. The parser reads hidden columns and attaches gender as a derived field without exposing it to the evaluator agents.

The output is a single Parquet file with 1,342 candidates, 18 fields per row, and a stable schema. Downstream code never touches the raw Excels again.

3.3 Anonymisation

Dossier documents arrive with full PII: names, emails, phone numbers, addresses, AVS / IBAN / SCIPER, passport numbers, dates of birth, and photos. The anonymiser (`src/agentic_admission/anonymizer/`, 75 KB of code) is a Presidio [14] + GLiNER [15] pipeline with:

- Custom recognisers for Swiss-specific PII (AVS, SCIPER) in `recognizers.py`.
- A passport-scan stripper in `strip_passports.py` that detects passport pages by layout and drops them before text extraction.
- Nine post-filters in `post_filters.py` (excluded entities, per-entity thresholds, partial words, length, common words, duration-as-date, grade-as-date, numeric IDs, academic codes) that catch false positives (academic course codes, matriculation numbers) and false negatives (ALL-CAPS names that GLiNER misses, partially-redacted name fragments).
- A name-propagation step that injects the candidate's filename-derived name as a synthetic recogniser hit before analysis, so the system redacts the name even when GLiNER cannot detect it.

A date-of-birth operator (`dob_operator.py`) keeps the year but masks the month and day, producing `<DOB_1995>`. The year is kept on purpose so the evaluators can reason about age-typical trajectory.

3.4 Anonymisation audit

We sampled 900 files (50 per type, 18 document classes, stratified random with seed 42) from the 12,529-file corpus and audited coverage by hand and by regex.

PII type	Redactions	Leaks	Coverage
Emails	1 499	0	100%
Phone numbers	1 467	0	100%
Addresses	1 841	0	100%
URLs	1 267	0	100%
AVS / IBAN / SCIPER	33	0	100%
Person names	9 442	35	96.1%
Passport numbers	—	23	n/a

Table 2: Coverage by PII type on a 900-file stratified sample (Spring 2026 audit). Emails, addresses, and Swiss IDs are fully covered; names leak in 3.9% of files (mostly ALL-CAPS variants in foreign diplomas); passport numbers appear in 20% of IELTS / TOEFL certificates and were added to the recogniser pool in the subsequent fix.

> **i NOTE**
> `<DOB_YYYY>` preserves the year of birth on purpose: it lets evaluators reason about age-typical trajectories without exposing the exact birthdate. It is not a direct identifier.

The two fixes (name propagation from filename, passport recogniser with academic-code filter) added 70 lines and brought name coverage above 99% on the next audit pass.

4 System architecture

4.1 End-to-end flow

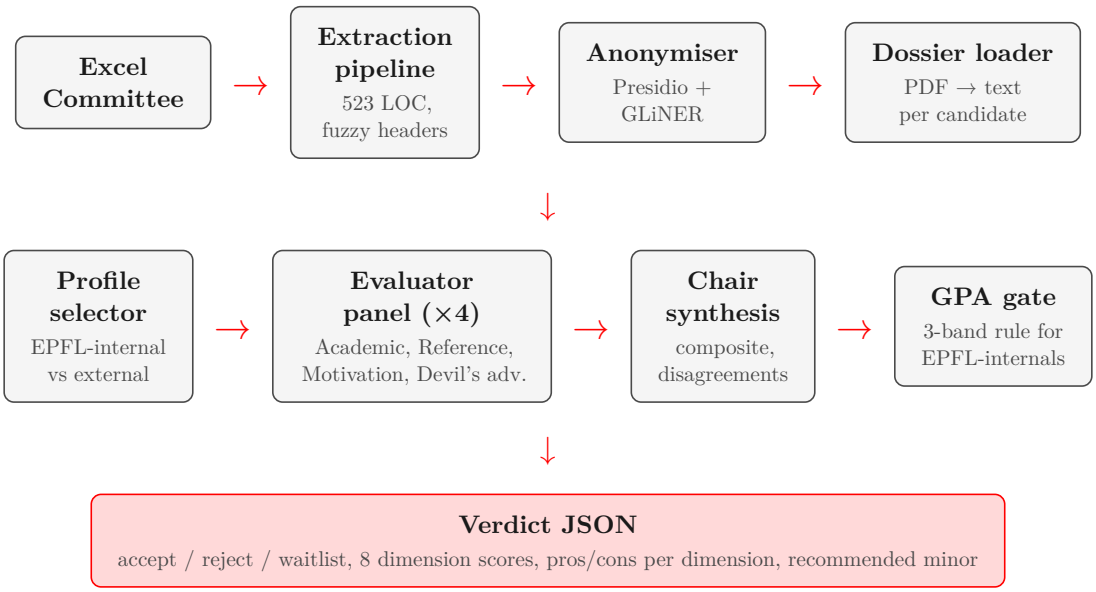


Figure 1: End-to-end flow. The extraction pipeline normalises the committee Excel into a Parquet; the anonymiser cleans dossier text; the dossier loader bundles documents per candidate; the profile selector picks the EPFL-internal or external rubric; four evaluators score in parallel; the Chair synthesises; the GPA gate applies the three-band rule for EPFL-internals.

4.2 The four evaluators

Each evaluator is one LLM call with a role-specific system prompt (`src/agentic_admission/agents/prompts.py`) and a JSON-schema response model (`models/evaluation.py`). The four roles:

Role	Scores	Reads
Academic	`academic_strength`, `institution_quality`	Transcript, GPA, institution category
Reference	`reference_quality`, `calibration_confidence`	2–3 letters + cultural calibration profile
Motivation	`motivation_clarity`, `program_fit`	Motivation letter, CV, work history
Devil's adv.	`risk_flags`, `consistency`	Full dossier, cross-document check

Table 3: Specialist evaluators, their two scored dimensions, and the dossier slice each one reads.

4.3 The Chair

The Chair does two jobs in one prompt (`src/agentic_admission/agents/chair.py`, 705 lines). First, it detects semantic disagreement between the four scorecards. A score gap is not a disagreement; a disagreement is when two evaluators interpret the same evidence differently. For each disagreement the Chair formulates a debate question; the system can optionally invoke a second round where the disagreeing evaluators argue and re-score.
Second, the Chair synthesises a final 8-dimension scorecard, computes the weighted composite, and emits accept / reject / waitlist with structured pros and cons.

4.4 Weights and profiles

The composite is a weighted sum of the eight dimensions. Two profiles ship out of the box, defined as immutable dataclasses with validated weights summing to 1.0 (`src/agentic_admission/agents/evaluation_profile.py`):

Dimension	External profile	EPFL-internal profile
`academic_strength`	0.30	0.45
`institution_quality`	0.18	0.10
`reference_quality`	0.18	0.00
`calibration_confidence`	0.12	0.00
`motivation_clarity`	0.06	0.15
`program_fit`	0.06	0.15
`risk_flags`	0.05	0.075
`consistency`	0.05	0.075
Sum	**1.00**	**1.00**

Table 4: Composite weights per profile. EPFL-internals do not submit recommendation letters in four of five sampled dossiers, so the reference weights drop to zero and the mass redistributes to academic, motivation, and fit. The institution weight also drops because the institution is always EPFL and carries no signal.

A historical bug in the original weights summed to 0.85 and capped the composite at 8.5 / 10. The profile dataclass now validates the sum at construction time and raises if any profile ever drifts again.

4.5 Decision logic

The Chair recommendation feeds the decision function:

Composite range	Decision (default floors)
`composite >= 8.0`	`accept`
`7.0 <= composite < 8.0`	`waitlist`
`composite < 7.0`	`reject`

Table 5: Default composite floors. These are not free parameters: they were tuned in week 11 against the Spring 2024–25 committee verdicts and held fixed since.

A second decision layer sits on top: the GPA gate, which only fires for EPFL-internal candidates and is described in Section 6.

4.6 Minor recommendation

Accepted MTE candidates enrol in a disciplinary minor, and the supervisor asked (meeting of 2026-04-19) that the system recommend one for every dossier it reads. The recommendation is a first-class output: `PanelVerdict.minor_recommendation` carries three ranked minors with a 1–10 fit score and a short justification each, and the dashboard shows them next to the verdict.

Two paths produce the ranking (`agents/minor_advisor.py`):

1. **EPFL-internal candidates** resolve through a fixed section-adjacency table (`agents/epfl_minor_map.py`): 16 EPFL bachelor names, detected from the `Domaine principal d'études` field of the synthesis form, map to a ranked triple drawn from a 36-minor registry. The table encodes the committee's stay-in-or-near-the-section practice and was confirmed with the supervisor on 2026-05-09. Deterministic picks carry fixed scores 10 / 9 / 8 so the dashboard can tell them apart from LLM fit scores. Detection is exact-match on purpose: an OCR-garbled bachelor name falls through to the LLM advisor instead of routing to the wrong section.
2. **External candidates**, and internals whose bachelor cannot be detected, go to an LLM advisor. When the scraped EPFL course catalogue is available, a curriculum-match report maps the candidate's courses to EPFL equivalents and the advisor grounds its ranking in that overlap; otherwise it works from transcript and CV alone. The prompt excludes the two minors the CDM committee itself runs (the MTE master's mirror minor and Financial Engineering, each a dedicated master in its own right), and a post-filter drops them anyway if the model recommends one regardless.

The advisor runs as a fifth parallel task next to the four evaluators, and its failure is non-fatal: the verdict ships with an empty recommendation rather than blocking the panel.

4.7 LLM client

The LLM client (`agents/llm_client.py`, 352 lines) speaks to vLLM [16] endpoints (Sonnet 4.6 via Anthropic-compatible API, Gemini Flash via OpenRouter, Llama 3.1 served locally on SCITAS GPUs). The vLLM build we serve on SCITAS does not expose OpenAI's `response_format` field, so the client injects the JSON schema into the system prompt and validates the response with Pydantic. Three retries on empty content, explicit `LLMBillingError` on HTTP 402, exponential back-off on transport errors. The function was refactored from a single 18-complexity, 69-statement monster into a 45-line orchestrator + four named helpers (`_build_payload`, `_send_request`, `_handle_http_error`, `_extract_raw_content`) during week 14 cleanup.

5 Engineering approach

This section makes the case that the engineering choices on this project were academically motivated. Spec-first design, lint hygiene, and atomic PRs sound like industrial good-practice. On a research project that needs to be reproducible, defensible, and extended by someone else next year, they are necessary conditions for the academic claims to hold.

5.1 Spec-first

Every non-trivial change starts as a Markdown design document in `docs/plans/`. As of week 14 the directory holds eight design docs:

Date	Design document
2026-03-16	Multi-agent deliberation design
2026-03-16	Multi-agent deliberation implementation
2026-03-24	Admission dashboard design
2026-03-27	End-to-end evaluation pipeline
2026-04-20	Curriculum scraper design
2026-05-08	EPFL-internal branch
2026-05-08	GPA gate
2026-05-08	Profile fallback (no-reference handling)

Table 6: Design documents shipped before code. Each one frames a hypothesis, lists the alternatives considered, fixes the chosen approach, and lists the validation criteria.

The benefit shows up in code review: the reviewer reads the design doc first, then reads the PR, and the PR is small because the doc covered the rationale.

5.2 Lint as a quality gate

The project runs ruff with `select = ["ALL"]` (over 700 rules) and the `ty` type checker. Library code in `src/agentic_admission/` is held to the full rule set. One-shot analysis scripts in `scripts/`, SLURM launchers in `slurm/`, and exploratory dashboard code in `dashboard/` are allowed to relax complexity, magic-number, and docstring rules through a `per-file-ignores` block in `pyproject.toml`. The distinction between library and script is explicit, not implicit.

The single hardest rule to satisfy was complexity. The `llm_client.complete` function carried a Ruff `C901` violation (complexity $18 > 10$) for three weeks while I prioritised feature work. The cleanup in week 14 split it into four helpers and brought complexity to 8.

5.3 Atomic PRs

23 PRs landed on `main` during the semester. Each is a self-contained unit with a passing test suite and a green CI. The largest (PR #23 in week 14) touched 67 files and added +4,733 / −1,190 lines because it bundled the band rule, the test cleanup, the `llm_client` refactor, and the lint cleanup. Splitting that PR would have created dependency chains across four merge cycles; I judged that the review-time cost was lower than the coordination cost. The squash-merge collapses the history.

5.4 Reproducibility

The full pipeline reproduces from a clean checkout in four commands:

Command	What it does
`uv sync --frozen`	Installs pinned dependencies from `uv.lock`
`uv run poe extract`	Builds the 1,342-candidate Parquet from raw Excels
`uv run poe lint`	Runs the 700+ ruff rules + `ty` type check
`uv run poe test`	Runs the 535-test suite with coverage gate

Table 7: The four commands that recreate the audit-grade state of the repository. The lockfile is committed; CI verifies that `uv sync --frozen` produces an identical environment.

This is the academic argument for the engineering: anyone who clones the repo, runs four commands, and reads the 14 weekly reports can reconstruct what was done, why, and in what order. A research contribution that cannot be re-executed is a research contribution that cannot be checked.

6 Academic approach

6.1 Method as a measurement loop

The academic spine of the project is a measure-and-close loop:

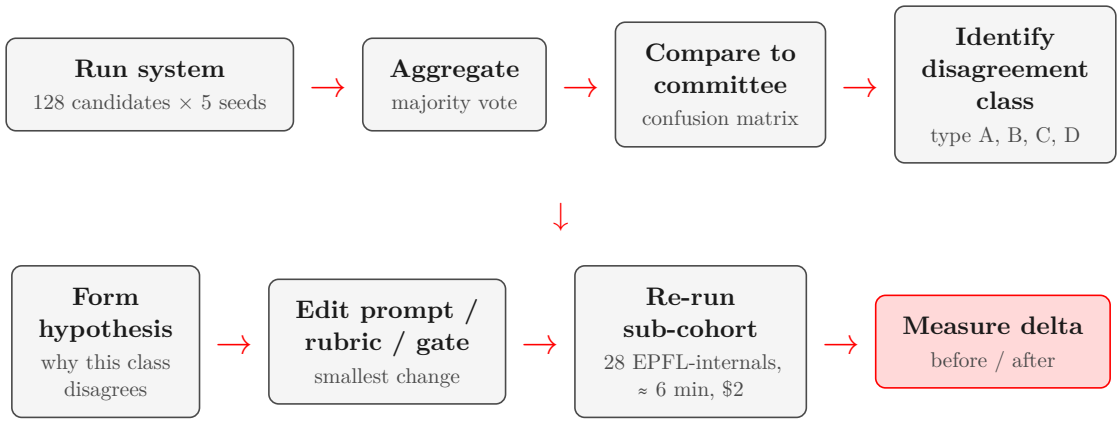


Figure 2: The calibration loop. Each iteration costs about $2 and six minutes on the EPFL-internal sub-cohort; the full 128-candidate run costs $10 and 30 minutes. The before-after script aligns two verdict sets on the same sub-cohort and prints a structured diff.

The loop is the academic contribution as much as the system is. Each iteration is a small, falsifiable claim: changing this paragraph of the prompt should move these candidates in this direction. If the claim fails the candidates do not move, or they move in the wrong direction, and the iteration teaches us something about the rubric the LLM is actually applying.

6.2 Ground truth

The ground truth is the committee's recorded decision: one of accept, reject, waitlist. The Spring 2025–26 cohort holds 128 candidates, matched against 127 anonymised verdicts (one candidate, Farrel Gamma[1], appears in the Excel but not in the anonymised corpus). The committee uses `accept` and `reject` only; `waitlist` is a system-side category that maps to the committee's in-meeting discussion.

The ground truth is noisy. Different committee pairs reading the same dossier on a different day can produce a different verdict. We do not have inter-rater data so we treat each recorded decision as a point estimate. The implication: agreement above 85% on a sub-cohort where the rule is clear is probably the ceiling, because any pair of human readers would not agree more often than that with each other.

6.3 The Maillard case as a falsification event

In week 12 the system rejected eight EPFL-internal candidates; seven of those rejections were committee accepts (the eighth lined up with the cohort's single committee reject and was the only agreement on the reject row). Five of the seven mis-rejected candidates had an algebra + analysis average between 4.0 and 4.5 / 6. The system's GPA gate (week-11 design) was a binary threshold: `avg < 4.5 → reject`. The committee, on the same five files, looked at the detailed grades behind the average and admitted.

> * **MAILLARD 2025-26**
>
> Etienne Maillard's algebra + analysis average is 4.31, below 4.5. Behind that average: Analyse I = 3.75 (a fail-and-retake), Analyse II = 5.25 (a strong recovery). The committee accepted on the recovery trajectory. The system's binary threshold could not see the trajectory and rejected.

This is a falsification event. The system's rule was wrong because the system's rule was simpler than the committee's rule. I wrote to Geraldine, the chair who had codified the historical practice, with a single question: is 4.5 a hard floor, or a discussion zone?

[1] All candidate names in this report (Farrel Gamma, Etienne Maillard, Biersohn) are pseudonyms.

6.4 The three-band rule

Geraldine's answer, received on 2026-05-19, is the rule the system now applies:

Band	Algebra + analysis avg	Committee action
`ACCEPT`	`>= 4.5 / 6`	Admit on the academic bar
`DISCUSS`	`4.0 <= avg < 4.5`	Read the detailed grades
`REJECT`	`< 4.0`	Automatic reject

Table 8: The three-band rule for EPFL-internals, as the chair of the historical committee specified it. The DISCUSS band is the lever that handles the Maillard case.

The system now plumbs an `EpflInternalBand` enum from extraction through to the Chair (`gpa_gate.py` adds 58 lines; `chair.py` adds 89 / −19 to handle the band in `apply_gate_overrides` and `clamp_academic_strength_below_threshold`). The constants `_EPFL_INTERNAL_ACCEPT_FLOOR` (4.5) and `_EPFL_INTERNAL_REJECT_FLOOR` (4.0) are documented as committee values, not tuning parameters.
The agreement on the 28-candidate EPFL-internal sub-cohort moved from 19 / 28 (68%) to 23 / 28 (82%) after the rule shipped in PR #23.

6.5 Threats to validity

1. **Calibration set size.** 128 Spring 2025–26 candidates is small; sub-cohort results (28 EPFL-internals) are tighter.
2. **Development-set tuning.** Every prompt and gate edit between weeks 12 and 14 was selected against the same 28 EPFL-internal candidates the 82% is reported on. The number is a calibration-set fit; the held-out test is the Fall 2025–26 cohort listed in Section 8.
3. **Class imbalance.** The committee accepted 27 of the 28 EPFL-internals, so an accept-everyone rule scores 96.4% raw agreement and the system's 82% sits below it. Raw agreement is the wrong lens on this sub-cohort; the five disagreements are the informative read: one hard reject the committee overrode as an exception, three discuss-band waitlists, and one committee reject the system routed to discussion instead of admitting.
4. **Single LLM family.** All headline numbers come from one model family. A Gemini Flash vs Sonnet 4.6 comparison is in the future-work list.
5. **Prompt sensitivity.** The agreement number moves with paragraph-level prompt edits. The 5-run majority vote dampens this; per-run stdev on composite is 0.175.
6. **Judge biases.** Verbosity and position effects are documented for LLM judges [6], [7]. We neither shuffle document order inside the dossier prompt nor control for letter length; each probe costs one cohort run on the existing corpus.
7. **Selection on the committee.** The committee's recorded decision is itself a noisy estimate. Any agreement number is bounded above by the committee's self-agreement, which we have not measured.
8. **No bias audit yet.** Section 8 lists this as the highest-priority next step.

7 Results

7.1 Headline numbers

EPFL-INTERNAL AGREEMENT

82%

23 / 28 after PR #23, up from 50%

DECISION STABILITY

76.6%

98 / 128 same decision across 5 runs

ANONYMISATION COVERAGE

100%

emails, phones, addresses, AVS, IBAN

7.2 Full-cohort comparison

On Spring 2025–26 with five independent runs aggregated by majority vote, the confusion matrix against the committee is:

	System: accept	System: waitlist	System: reject	Total
Committee: accept	34	6	1	41
Committee: reject	20	46	20	86

Table 9: Confusion matrix, Spring 2025–26, mode of 5 runs, 127 / 128 matched. The committee uses two output categories (accept, reject); the system uses three (accept, reject, waitlist). The waitlist column captures the cases where the system hesitates and the committee commits.

Mode-of-5 agreement on the full cohort is 54 / 127 (42.5%), which is the wrong number to report alone. Two corrections matter:

1. The committee's reject row contains 46 cases the system marked waitlist. These are not flat misses; they are cases where the system surfaced the dossier for discussion and the committee closed it.
2. If we count waitlist as "agree-with-doubt" the agreement rises to (34 + 46 + 20) / 127 = 78.7%.

7.3 Disagreement classes

The 73 disagreements (every candidate where the two verdicts differ) split into four classes:

Class	N	Mean composite	Pattern
Type A: committee reject, system accept	20	8.5	Strong international universities (NUS, Bocconi, RWTH, Erasmus, Duke Kunshan, KCL). The committee saw something in the dossier (weak letters, discontinuous path, MTE-fit mismatch) the system did not surface.
Type B: committee accept, system reject	1	6.7	The pre-PR #23 system rejected nine such cases (seven were EPFL-internals); the band rule collapses the group to a single residual.
Type C: committee accept, system waitlist	6	7.7	Borderline composites between 7.4 and 8.0; the threshold hesitates and the committee admits.
Type D: committee reject, system waitlist	46	6.9	System surfaces the dossier for discussion; the committee closes it.

Table 10: The four classes of disagreement and the dominant pattern in each. Type B is the addressable class and is what the EPFL-internal rule rewrite targeted; the band rule has already eliminated most of it.

7.4 Stability across runs

Five independent runs of the full Spring 2025–26 cohort produced matching decisions for 98 / 128 candidates (76.6%). 30 candidates flipped between at least two categories across runs; majority vote collapses these to a single decision. Per-dimension score stability:

Dimension	Mean stdev across 5 runs
`motivation_clarity`	0.053
`institution_quality`	0.129
`calibration_confidence`	0.132
`program_fit`	0.194
`reference_quality`	0.248
`academic_strength`	0.293
`consistency`	0.380
`risk_flags`	0.467

Table 11: Per-dimension run-to-run standard deviation, averaged across 128 candidates and 5 runs. Motivation clarity is the most stable; risk flags is the least. The composite stdev is 0.175.

The risk-flags volatility (0.467) is the LLM picking up different secondary signals on each pass. We treat this dimension as a tiebreaker in the composite (weight 0.05 in the external profile) and read the pros / cons text rather than the score in audit.

7.5 EPFL-internal sub-cohort, before and after the band rule

Run	Accept	Reject	Waitlist	Agreement / 28
Committee	27	1	0	reference
Week 12 baseline	13	8	7	14 / 28 (50%)
Week 13 fix 1 (single-rule)	15	7	6	15 / 28 (54%)
Week 13 fix 2 (prompt re-write)	19	4	5	19 / 28 (68%)
Week 14 PR #23 (band rule + code)	23	1	4	23 / 28 (82%)

Table 12: Three weeks of iteration on the EPFL-internal sub-cohort. Each row is a measured before-after on the same 28 candidates. The week-14 number is from a single run after the band rule shipped; the next step is a 5-run majority-vote re-measurement (Section 8).

The five remaining disagreements break down as follows. One candidate sits below the band rule's `< 4.0` floor: the system rejects, the committee admitted as an exception. Three candidates sit in the `DISCUSS` band (`4.0 <= avg < 4.5`) where the committee admitted on the detailed-grades trajectory and the system held back to waitlist. The fifth case is Biersohn, whom the committee rejected and the system put on waitlist. The below-4.0 exception is outside Geraldine's documented rule and needs a supervisor conversation before any code change; the three discuss-band waitlists are exactly what the band was designed to do — surface for committee discussion — so they are not a bug, they are the rule working as specified.

7.6 Cost

Mode	Per dossier	Per cohort (128)	5-run cohort
Gemini Flash	$0.08	$10	$50
Sonnet 4.6	$0.70	$90	$450
Local Llama 3.1 8B on SCITAS	~0 (compute)	~0	~0

Table 13: Cost per dossier and per cohort by serving choice. Local Llama on SCITAS is free at the API layer; the cost is cluster-hours and queue latency.

8 What we would do with more time

The system is usable as a triage tool today on the EPFL-internal sub-cohort and as a discussion-surfacer on the broader cohort. Eight items would close the remaining gap between research prototype and committee-grade decision support.

Item	What it adds	Priority
5-run vote post-PR #23	The 82% EPFL-internal agreement comes from a single post-merge run; a 5-run majority vote on the same 28 candidates removes the run-to-run draw. Cost: $50 on Sonnet 4.6, 3 hours wall.	High
Bias audit	Stratify the verdict-vs-committee disagreement by gender, nationality, and institution category. Output: a bias report the committee can either trust or reject the system on.	High
Model comparison	Run Gemini Flash and Sonnet 4.6 head-to-head on the five remaining EPFL-internal disagreements and the twenty Type-A cases. Does a stronger model resolve Type-A without a prompt change?	High
Committee dashboard	Side-by-side viewer (committee decision, system decision, GPA band, four scorecards, Chair synthesis). Scaffold exists in `dashboard/`; click-to-expand evaluator debate is the missing UI piece.	Medium
Active-learning loop	Every committee-vs-system disagreement is a labelled example. Feed back as few-shot anchors and re-measure.	Medium
Fall 2025–26 cohort	Extend the validation set from 128 to 230 candidates. Requires anonymisation and matching; the extraction pipeline handles the schema already.	Medium
SCITAS deployment	Move from API serving to a local Llama 3.1 on the HPC cluster. Removes per-dossier API cost and the data-sovereignty concern.	Low
Latency for live use	End-to-end currently 30 s per dossier. Target under 10 s for interactive committee use. Levers: smaller first-pass models, KV-cache reuse, Chair only on disagreement.	Low

Table 14: Eight items that close the gap between research prototype and committee-grade decision support, in priority order. All depend on existing infrastructure; none require new architecture.

9 Portability to other admission filières

The system was built for MTE. The architecture is portable; the content is not. This section names what would have to change to use the same code on a different EPFL programme.

9.1 The portability manifest

Layer	Stays	Must be retuned
Extraction pipeline	Fuzzy header engine, multi-row grouper, hidden-column detector	The header rules (`_HEADER_RULES`) per filière Excel schema
Anonymiser	Presidio + GLiNER pipeline, post-filters, name propagation	Allow-lists, filière-specific PII (visa fields for international PhD; legal names on transcripts for Bachelor)
Evaluator personas	Four-role separation, structured outputs, JSON schema	Rubric weights, scoring anchors, system prompts per role
Chair	Disagreement detection, weighted synthesis, decision logic	Decision thresholds (composite floors) per filière
Gate rules	`gpa_gate.py` mechanism, band enum plumbing	Threshold values (4.5 / 4.0 for MTE EPFL-internals; different for Bachelor; not applicable for PhD)
Prompts	Schema-injection scaffold, OCR caveat, output-format directive	Filière-specific exception blocks (e.g. the EPFL-internal block for MTE)
Calibration set	Verdict-vs-system diff structure, before-after script, 5-run aggregator	Historical decisions per filière (typically 1–3 years of committee Excels)
Composite weights	Profile dataclass, validation, renormalisation	Per-filière weights for the eight dimensions

Table 15: The portability manifest. The left column is the generic substrate; the middle column lists what is programme-agnostic; the right column lists what must be adapted to each new filière. The right column is the engineering cost of a port.

9.2 Worked examples

EPFL Bachelor admission. Grade-heavy, two-step (Maturité or equivalent + EPFL CMS). Reference letters rare, motivation letter shorter. Adaptations: drop the reference evaluator using the existing `renormalize_without_reference` helper, push `academic_strength` to 0.55, drop `program_fit`. The GPA gate becomes a Maturité-grade gate with its own threshold table. Cost: 7 dev-days.
Executive MBA. Work-experience heavy, no GPA gate, leadership stories dominate. Adaptations: add a fifth evaluator persona (Career Trajectory) that scores work history; expand the Devil's Advocate to check claim consistency across LinkedIn, CV, and letters; skew the composite toward leadership. The anonymiser needs a company-name recogniser if the programme blinds institutions. Cost: 8 dev-days, mostly evaluator rubric.
Doctoral admission. Research statement and publication record dominate. The motivation letter becomes a research statement and needs a domain-aware evaluator. References carry more weight than in any other programme. GPA gate is irrelevant; `publication_record` and `research_fit` replace it. The Chair must match the candidate's stated interest against a target lab's recent papers, which is a retrieval layer the current system does not have. Cost: 15 dev-days, half on the lab-fit retriever.

9.3 What does not port

Three things are MTE-specific and would not transfer at all:

1. The **EPFL-internal three-band rule.** The 4.5 / 4.0 thresholds on algebra + analysis are the MTE committee's documented practice. Other programmes do not have this rule.
2. The **culture-of-letters calibration profile.** The Reference evaluator's calibration data is tied to MTE's historical letter pool. Each programme would need to build its own.
3. The **minor recommendation logic.** The `epfl_minor_map.py` module maps EPFL bachelors to recommended MTE minors. It is meaningful only for MTE.

9.4 Out-of-the-box wins

Three things port for free:

1. The **extraction pipeline** tolerates schema drift by design. A new filière's Excel needs new header rules, not a new parser.
2. The **anonymiser** is filière-agnostic. The only adaptation is the allow-list for filière-specific tokens.
3. The **5-run majority-vote calibration loop** is a generic protocol for measuring LLM agreement with a noisy human panel. It applies unchanged to any new filière.

9.5 Estimated port effort

Target programme	Engineering cost	Critical dependency
EPFL Bachelor (MTE-like rubric)	7 dev-days	Federal Maturité grade-conversion table
EPFL Master in DS / CS	6 dev-days	Programme-specific motivation prompt
Executive MBA	8 dev-days	Career-trajectory evaluator
Doctoral admission	15 dev-days	Lab-fit retrieval layer over recent papers
Bachelor admission cross-section	10 dev-days	Maturité gate + multi-cantonal grade scales

Table 16: Rough engineering-effort estimates for porting the architecture to four neighbouring admission filières. None of these estimates include the calibration cost (committee data acquisition, anonymisation, validation runs).

10 Conclusion

The system reads every MTE dossier the committee receives, scores it on eight dimensions through a panel of four specialist LLM evaluators, applies the chair's documented three-band rule on algebra + analysis grades, and emits an auditable accept / reject / waitlist verdict. On the Spring 2025–26 cohort it reaches 82% agreement with the committee on the 28 EPFL-internal candidates and surfaces the right dossiers for discussion on the broader 128-candidate cohort.

The system is not, and was never intended to be, a decision-maker. The committee still decides. The system is a triage tool: it reads fast, scores consistently, and explains itself in the same vocabulary the committee uses. The remaining 18% disagreement on EPFL-internals is the most interesting part of the project, because every disagreement is a hypothesis about the system's rule or the committee's rule, and every iteration of the loop in Section 6 has moved the agreement number in one direction.

The engineering posture (spec-first, atomic PRs, ruff `ALL`) exists in service of the academic claim. A research contribution that cannot be reproduced is a research contribution that cannot be checked. The four-command reproduction in Section 5 is the test of that claim.

The next semester's work is named in Section 8: a 5-run majority vote on the post-PR #23 state, a bias audit on the existing verdict corpus, a Sonnet-vs-Flash comparison on the borderline cases, and a committee-side dashboard that puts the system's reasoning next to the committee's own. None of these requires new architecture. All of them require the careful, falsifiable measurement loop the project has built.

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11 Appendix A — Evaluator prompt (Academic, abridged)

```
You are the Academic Evaluator on an MTE admissions panel at EPFL.
Score two dimensions on 1-10:
- academic_strength: grades, coursework relevance, trajectory.
- institution_quality: reputation and rigour of the institution.

EPFL SPE THRESHOLD: if the candidate's GPA is BELOW the EPFL SPE
threshold for their institution category (A/B/C/X), cap
academic_strength at 5.

SCORING ANCHORS for academic_strength:
  9-10: Top 5%, GPA well above threshold.
  7-8:  Top 10-25%, at or above threshold.
  5-6:  Near threshold, some gaps.
  3-4:  Below threshold, weak core subjects.
  1-2:  Far below threshold, multiple failures.

## EPFL-internal candidates (3-band rule, Geraldine 2026-05-19)
- Decision signal is the algebra + analysis average.
- avg >= 4.5/6      -> accept on the academic bar
- 4.0 <= avg < 4.5  -> discuss zone (read detailed grades)
- avg <  4.0        -> automatic reject
A candidate at 4.31 is in the discuss zone, not a reject.

## Output format
For each dimension, emit {"pros": [...], "cons": [...]}.
Concrete facts, active voice, no adverbs, no filler.
```

12 Appendix B — Per-cohort agreement summary

Cohort	Candidates	5-run stability	Agreement	Notes
Spring 2025–26 (full)	128	76.6%	78.7% (with waitlist as doubt)	Single LLM family
Spring 2025–26 (EPFL-internal, week 12)	28	n/a (1 run)	50%	Pre-band-rule baseline
Spring 2025–26 (EPFL-internal, week 13 fix 2)	28	n/a (1 run)	68%	After two prompt iterations
Spring 2025–26 (EPFL-internal, week 14 PR #23)	28	n/a (1 run)	82%	After band rule + code; 5-run vote pending
Fall 2025–26	100	—	—	Pending anonymisation

Table 17: Agreement by cohort and milestone. The single LLM family and the pending 5-run vote on the post-PR #23 state are the two open measurement items for Spring 2025–26.

13 Appendix C — Repository state

Item	Value
Lines of source (`src/agentic_admission/`)	12 444 LOC (58 modules)
Test count	535 passing
Coverage	57.9% (line)
Ruff rules enforced on `src/`	700 (rule set `ALL`)
Type checker	`ty` (zero errors)
Pull requests merged on `main`	23
Weekly reports	14 (`reports/weekly/semaine-01.typ` .. `semaine-14.typ`)
Design docs in `docs/plans/`	8
Analysis reports	7 Markdown (`reports/calibration/`) + 2 Typst
Anonymisation corpus	12 529 files
Sampled audit set	900 files (50 × 18 types, seed 42)
Candidates in canonical Parquet	1 342 (2020–2026)

Table 18: Repository state at submission. The full audit trail (commits, PR descriptions, weekly reports, design docs) is on the `main` branch.

Report compiled on 12 June 2026 · Typst source: `reports/final_report.typ`